

Lab 01: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Name: Malaika Irfan

Registration Number: FA24B1-SE-004

Course Code: SE-665

Section: B

GitHub Username: MalaikaIrfan004

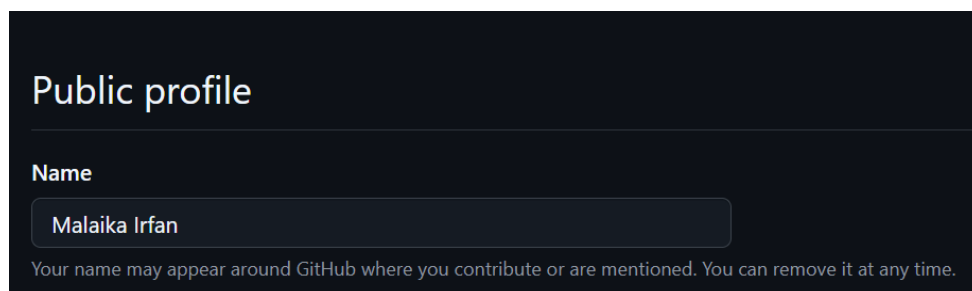
Task 1: Create and Configure Your GitHub Account

1. github_profile.png



GitHub profile showing the user's actual name and username

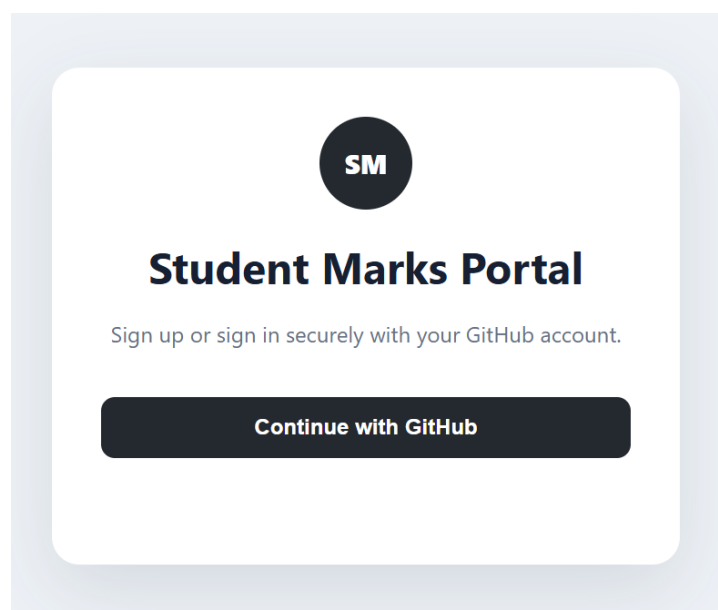
2. github_actual_name.png



GitHub Public Profile showing the user's complete actual name.

Task 2: Register in the Student Marks Portal

3. portal_login_page.png



Student Marks Portal login page showing the available GitHub sign-in option.

4. portal_github_profile.png



Student Marks Portal showing the authenticated GitHub profile information.

5. portal_enrollment_submitted.png

ID

Complete registration

Enter your registration number, course and section. The enrollment requires approval.

Registration number

2024-BSE-04

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

SE-665 — Cloud Computing

Section

Section B

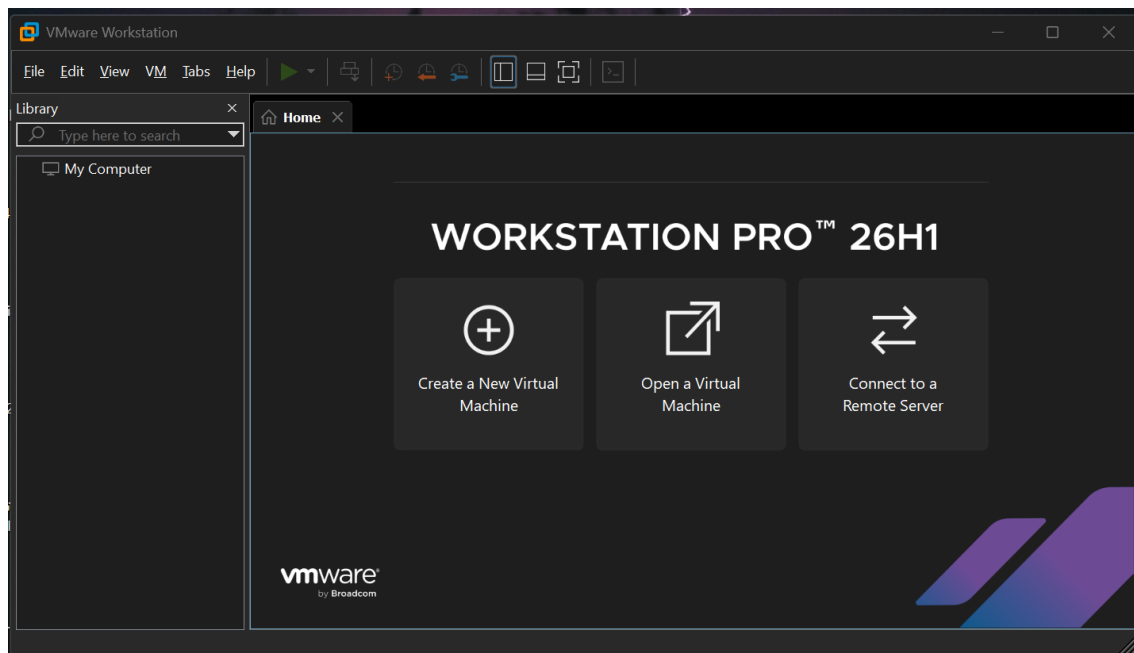
Submit enrollment

Sign out

Student Marks Portal showing the submitted enrollment and its current status.

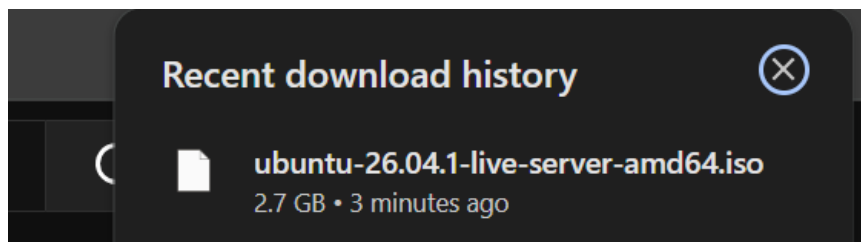
Task 3: Download VMware Workstation Pro and Ubuntu Server

6. vmware_workstation.png



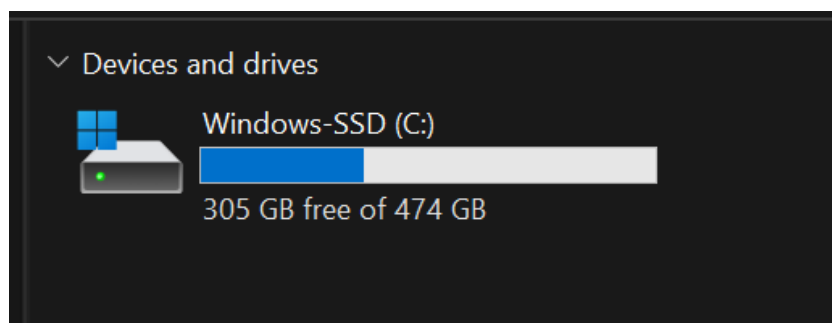
VMware Workstation Pro running successfully on the Windows computer.

7. ubuntu_server_iso.png



Downloaded Ubuntu Server LTS ISO file shown with its filename.

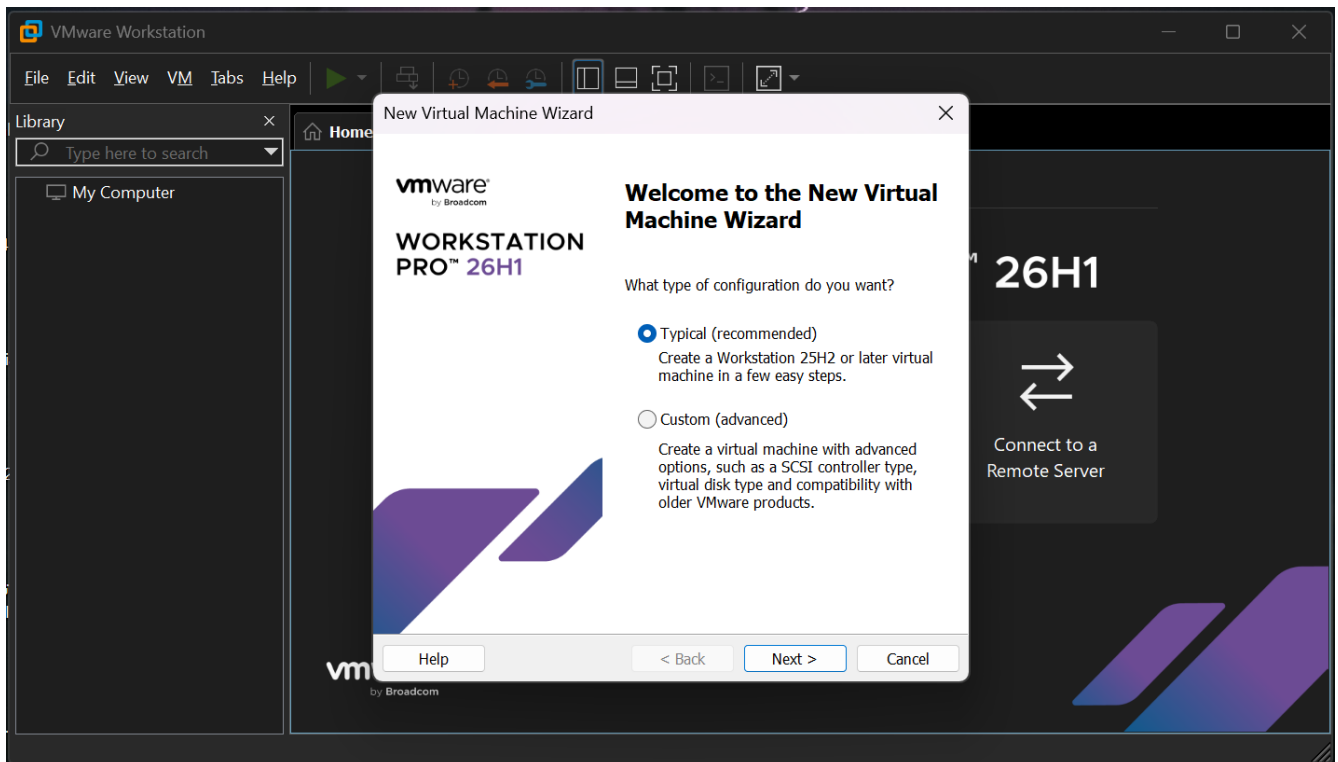
8. available_storage.png



Windows storage showing at least 20 GB of available free space.

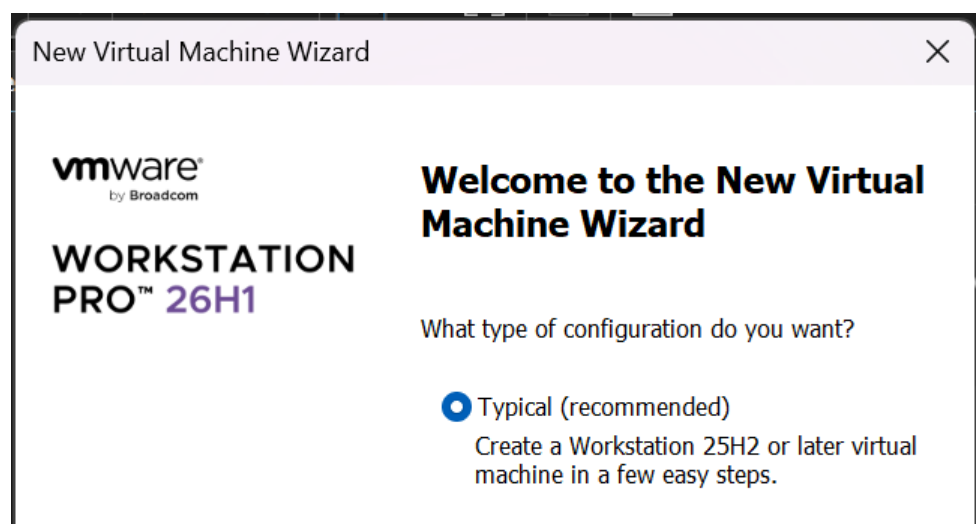
Task 4: Create the Ubuntu Server Virtual Machine

9. vm_creation_wizard.png



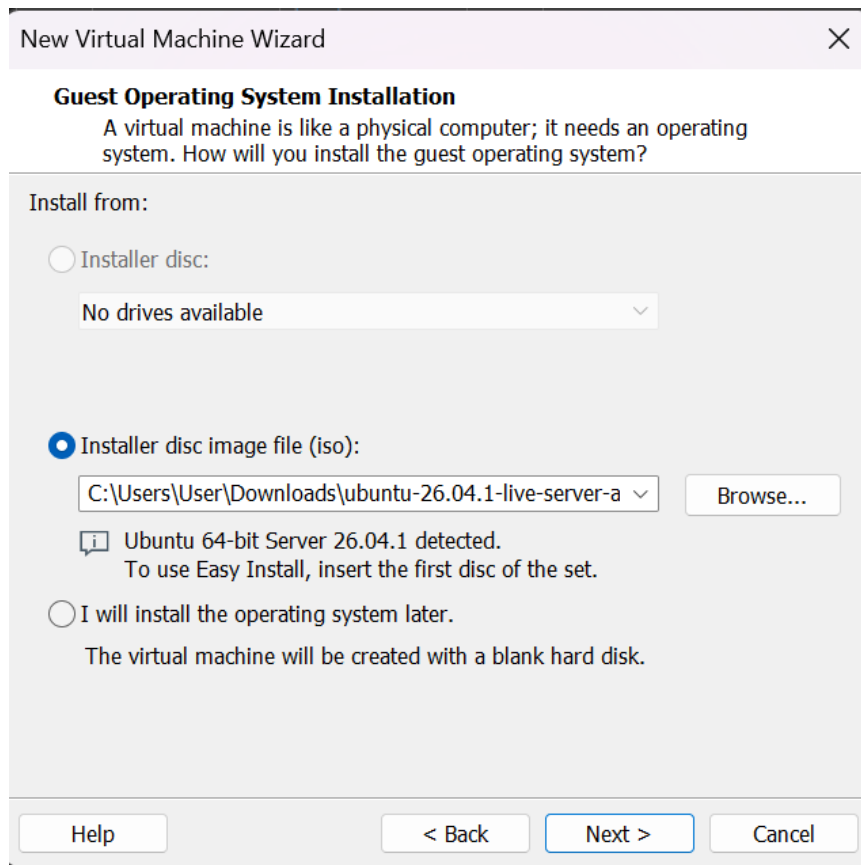
VMware New Virtual Machine Wizard opened to create the Ubuntu Server VM.

10. vm_typical_configuration.png



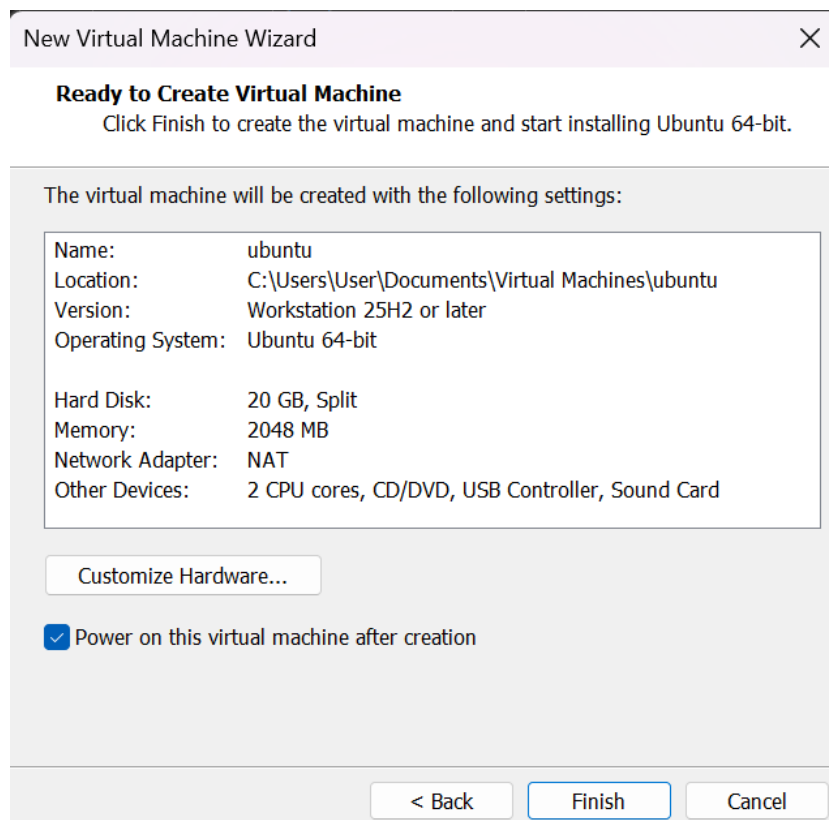
VMware wizard showing the Typical recommended configuration selected.

11. vm_ubuntu_iso_selected.png



Ubuntu Server ISO selected as the virtual machine installation media.

12. vm_configuration_summary.png



Final VMware virtual machine configuration before creation.

13. ubuntu_installer_boot.png

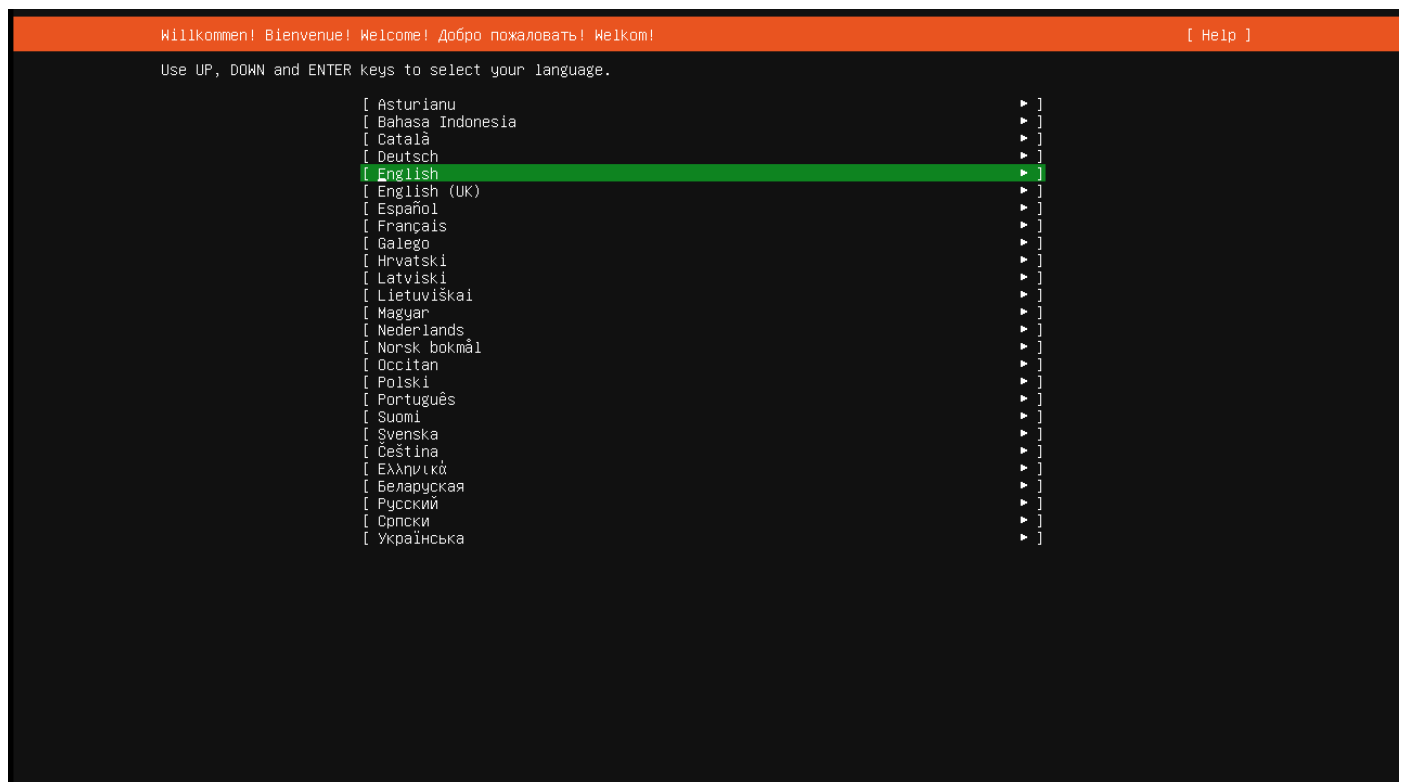
```
[ OK ] Finished initrd-cleanup.service - Cleaning Up and Shutting Down Daemons.
[ OK ] Stopped systemd-networkd.service - Network Management.
[ OK ] Closed systemd-networkd-resolve-hook.socket - Network Management Resolve Hook Socket.
[ OK ] Stopped target network-pre.target - Preparation for Network.
[ OK ] Closed systemd-networkd.socket - Network Management Netlink Socket.
[ OK ] Stopped systemd-sysctl.service - Apply Kernel Variables.
[ OK ] Stopped systemd-modules-load.service - Load Kernel Modules.
      Stopping systemd-udevd.service - Manager for Device Events and Files...
[ OK ] Stopped systemd-udevd.service - Manager for Device Events and Files.
[ OK ] Closed systemd-udevd-control.socket - udev Control Socket.
[ OK ] Stopped dracut-pre-udev.service - dracut pre-udev hook.
[ OK ] Stopped dracut-cmdline.service - dracut cmdline hook.
[ OK ] Stopped dracut-cmdline-ask.service - dracut cmdline hook.
      Starting initrd-udevadm-cleanup-db.service - Cleanup udev Database...
[ OK ] Stopped systemd-network-generator.service - Generate Network Units from Kernel Command Line.
[ OK ] Stopped systemd-tmpfiles-setup-dev.service - Create Static Device Nodes in /dev.
[ OK ] Stopped systemd-tmpfiles-setup-dev.service - Create Static Device Nodes in /dev gracefully.
[ OK ] Stopped kmod-static-nodes.service - Create List of Static Device Nodes.
[ OK ] Finished initrd-udevadm-cleanup-db.service - Cleanup udev Database.
[ OK ] Reached target initrd-switch-root.target - Switch Root.
      Starting initrd-switch-root.service - Switch Root...
```

Welcome to Ubuntu 26.04.1 LTS!

Ubuntu Server installer boot screen displayed after powering on the VM.

Task 5: Choose the Installation Language

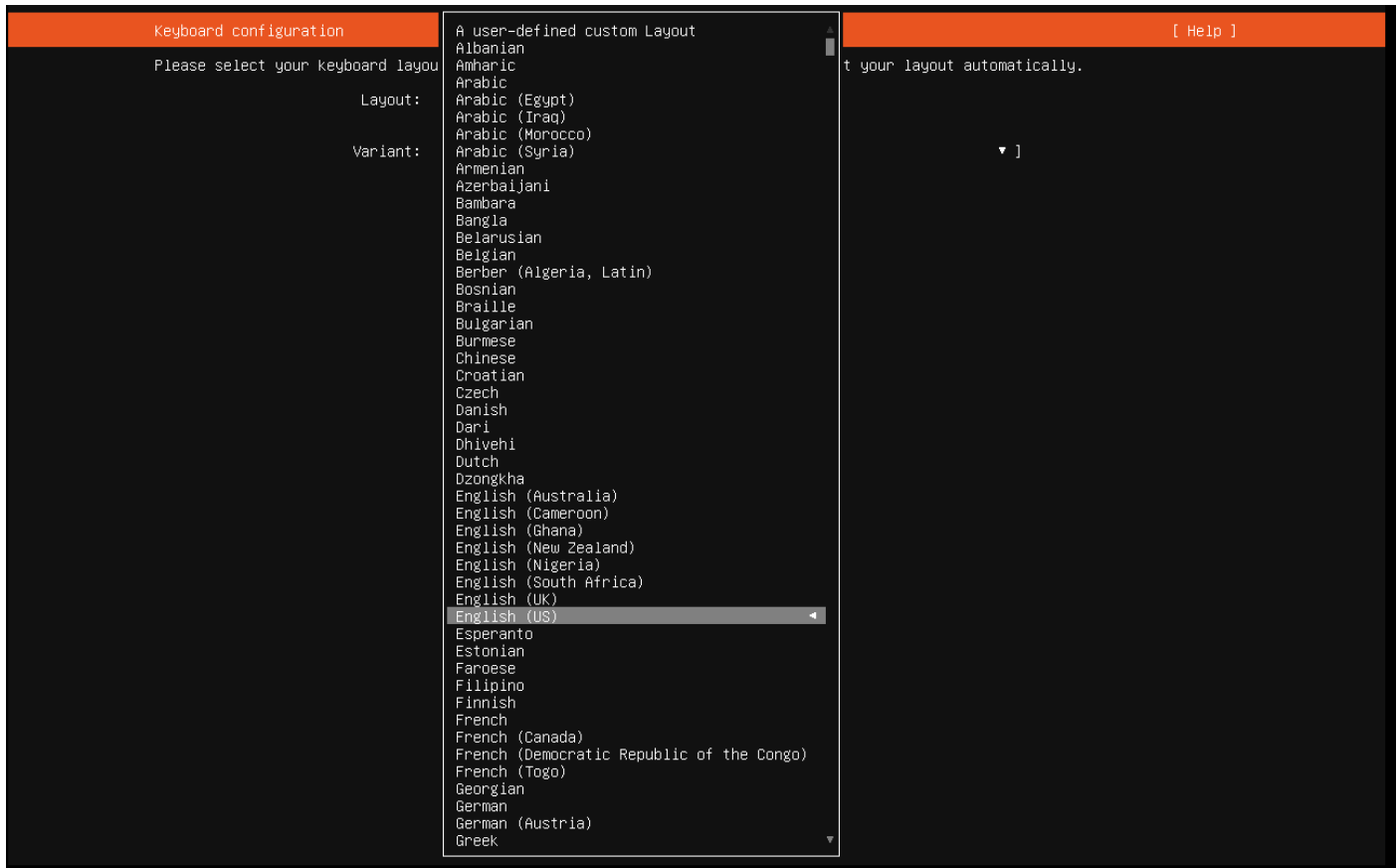
14. ubuntu_language.png



Ubuntu Server installer showing the selected installation language.

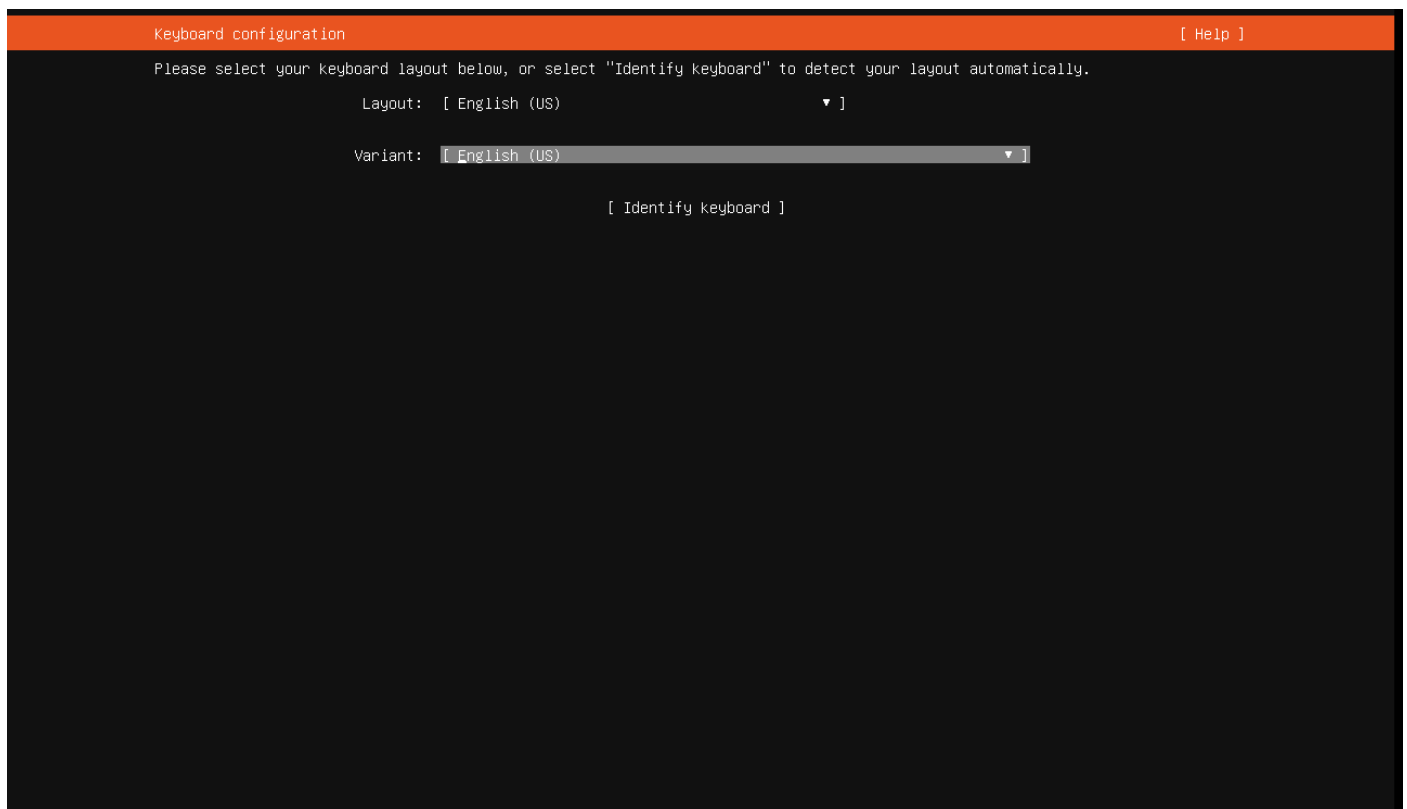
Task 6: Configure the Keyboard Layout

15. ubuntu_keyboard_layout.png



Ubuntu installer showing the selected keyboard layout.

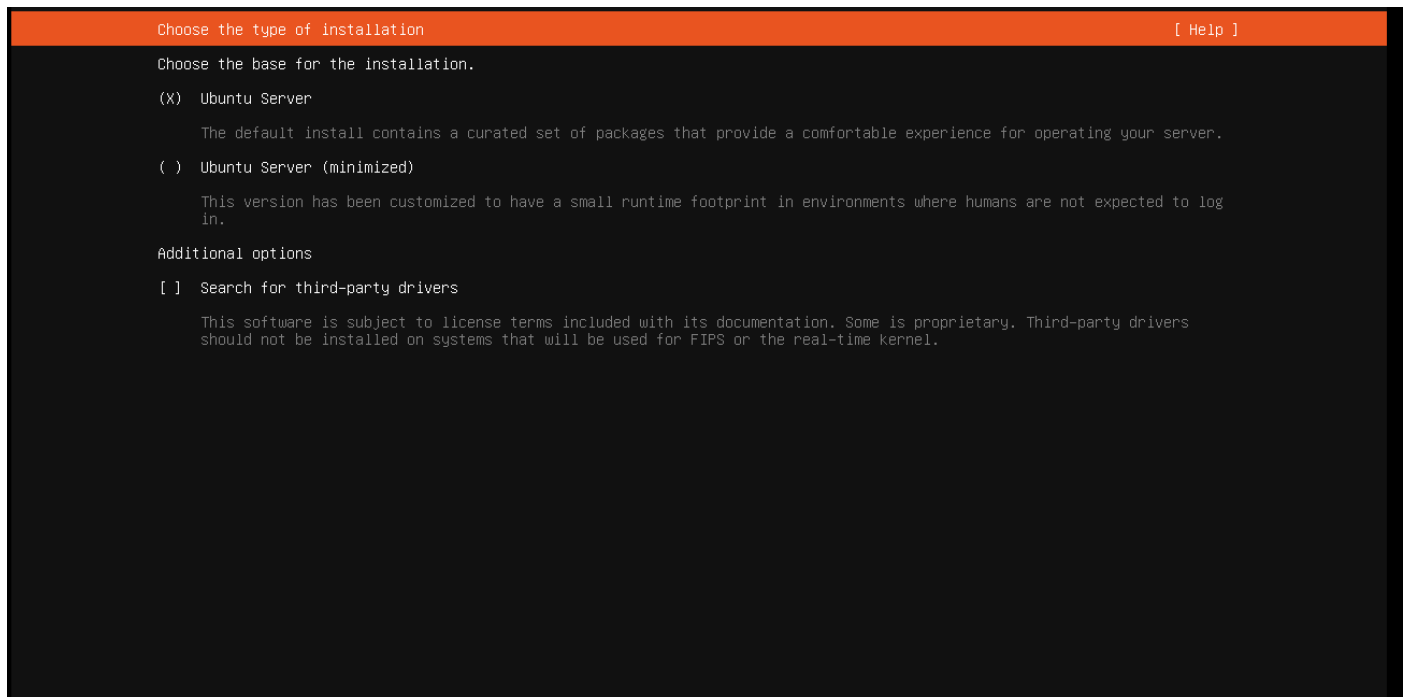
16. ubuntu_keyboard_variant.png



Ubuntu installer showing the selected keyboard variant.

Task 7: Choose the Ubuntu Server Installation Type

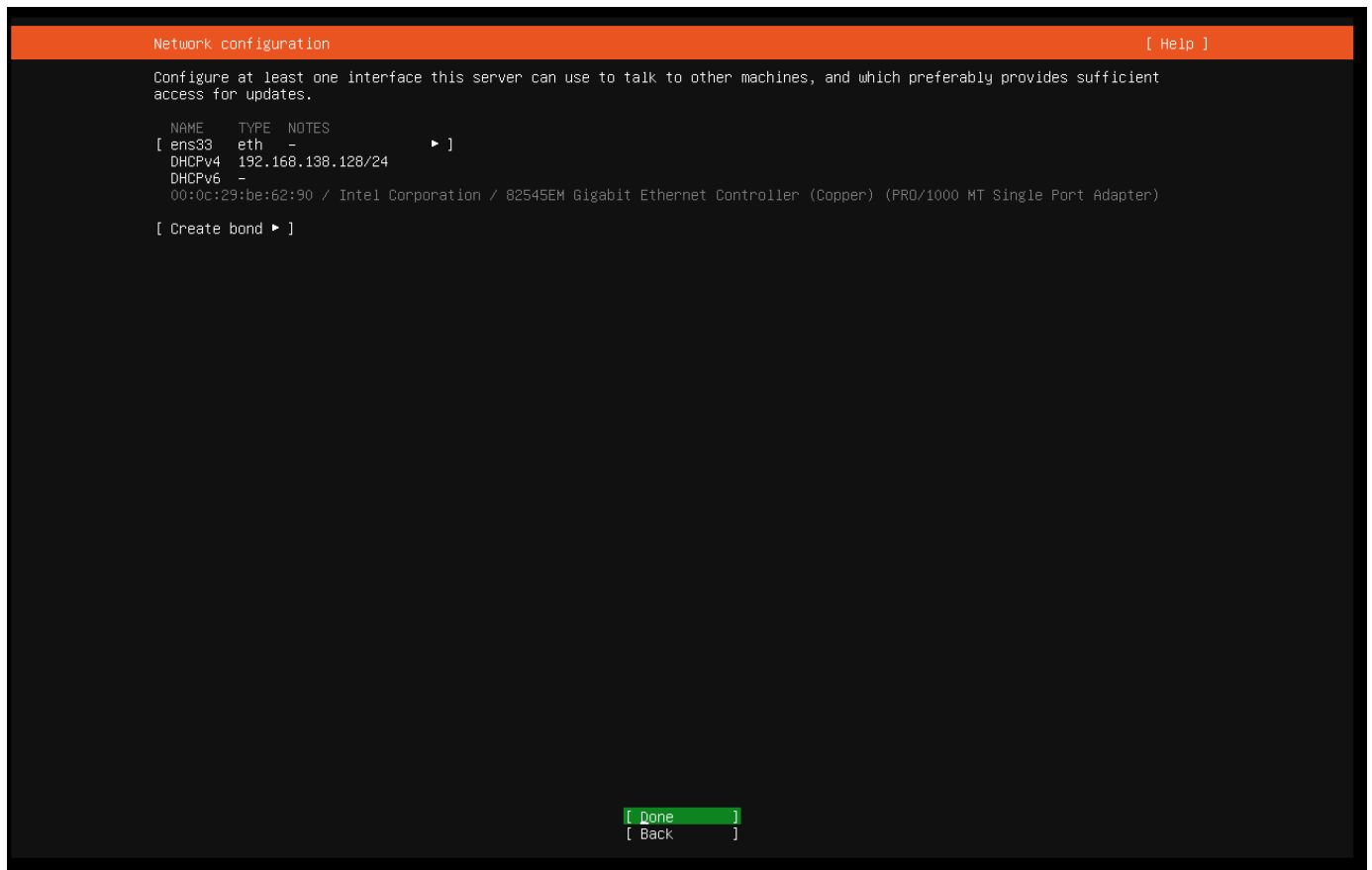
17. ubuntu_installation_type.png



Ubuntu Server installer showing the normal server installation option selected.

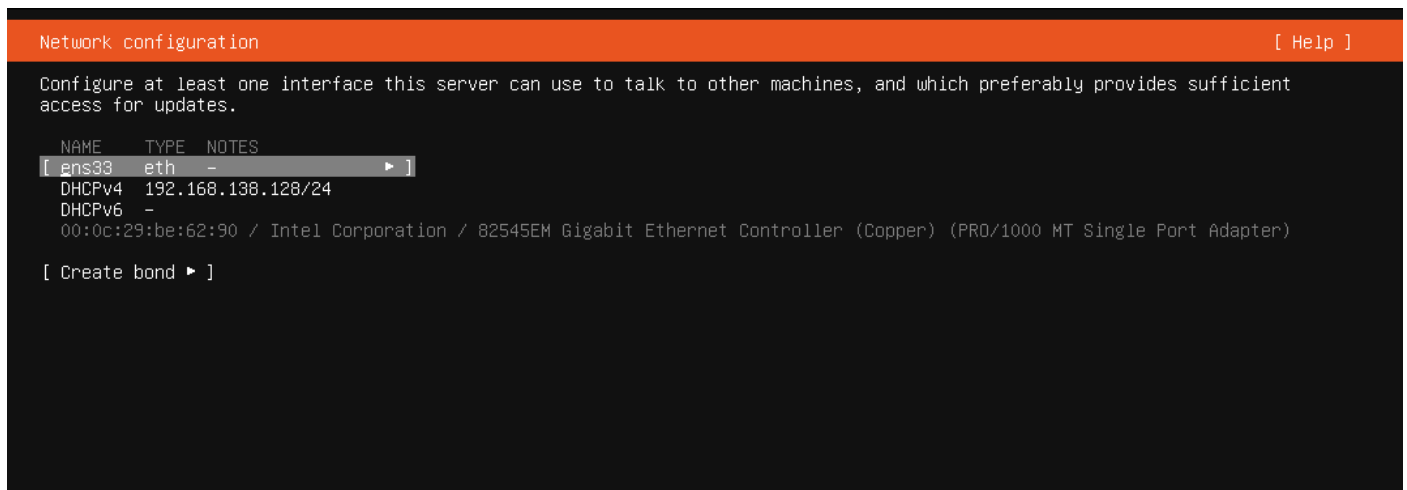
Task 8: Configure the Network

18. ubuntu_network_interface.png



Ubuntu installer showing the detected network interface.

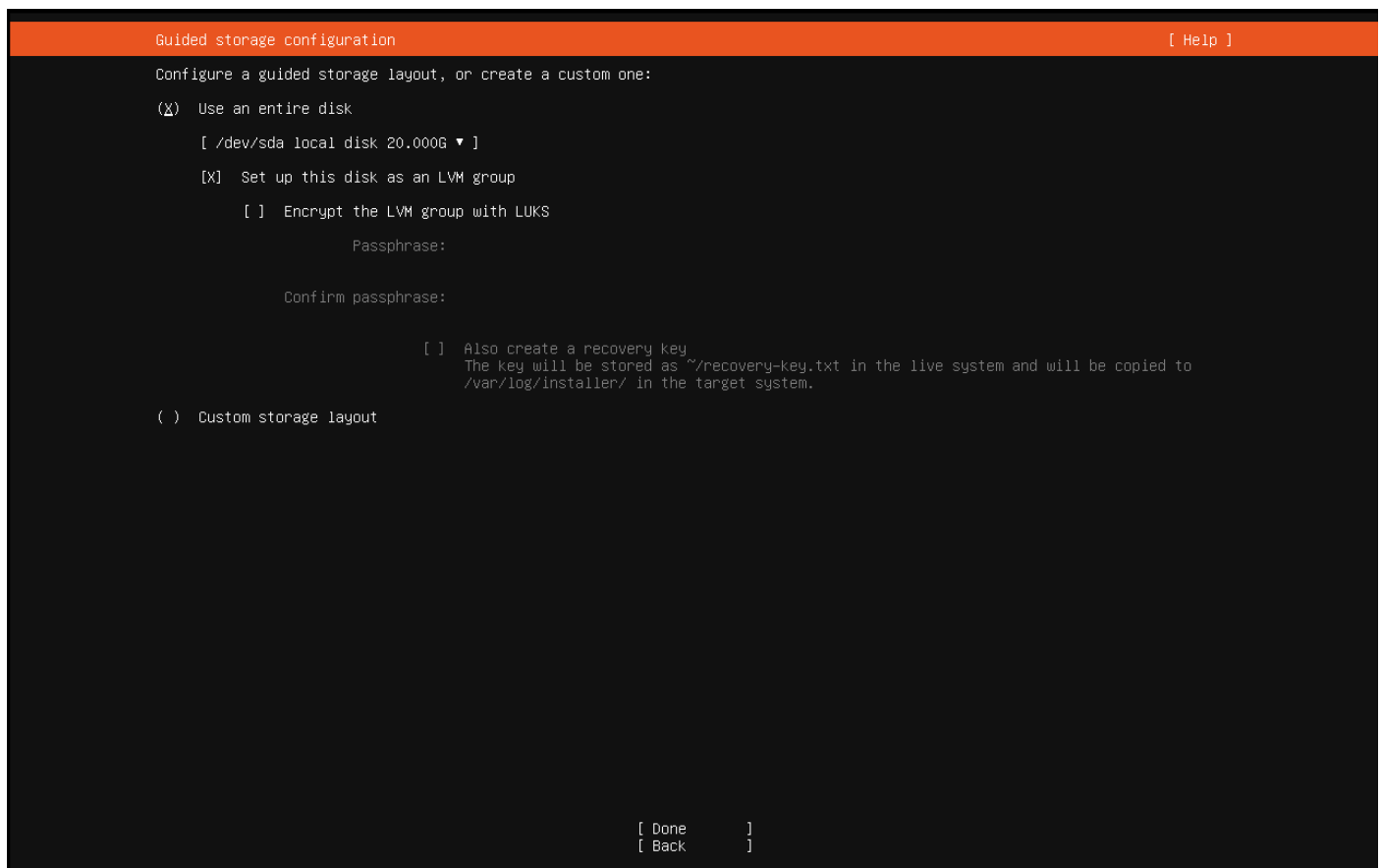
19. ubuntu_installer_network.png



Ubuntu installer showing the IPv4 address assigned through DHCP.

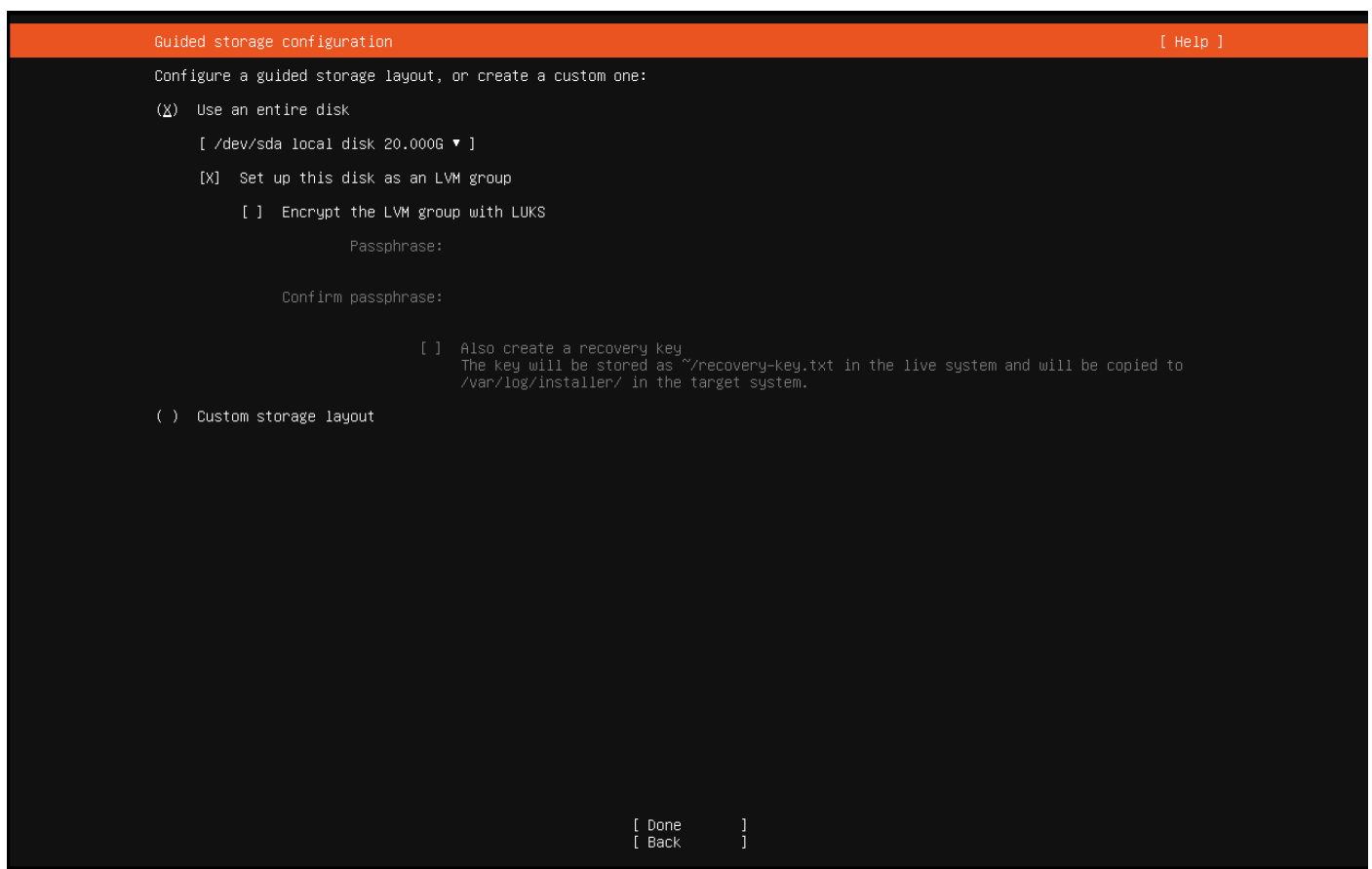
Task 9: Configure Storage

20. ubuntu_guided_storage.png



Ubuntu installer showing the guided storage option selected for the VMware disk.

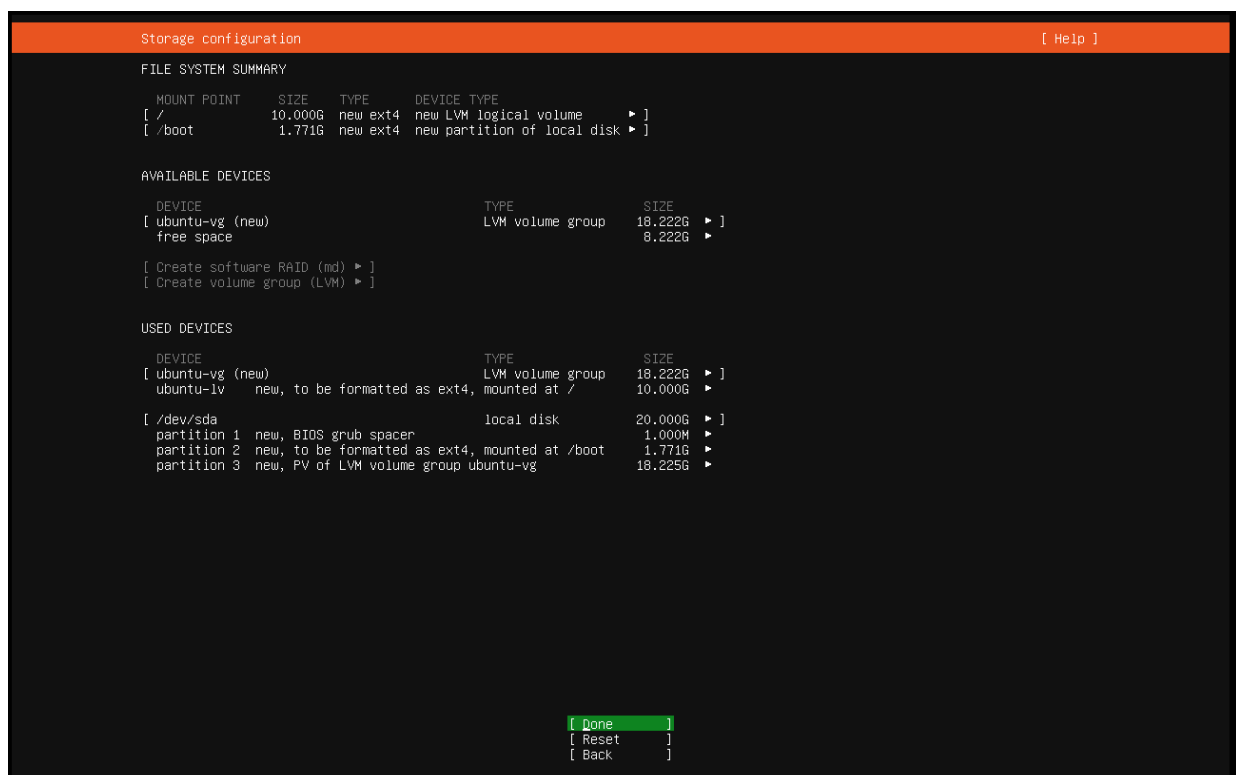
21. ubuntu_storage_configuration.png



Ubuntu installer showing the proposed storage configuration for review.

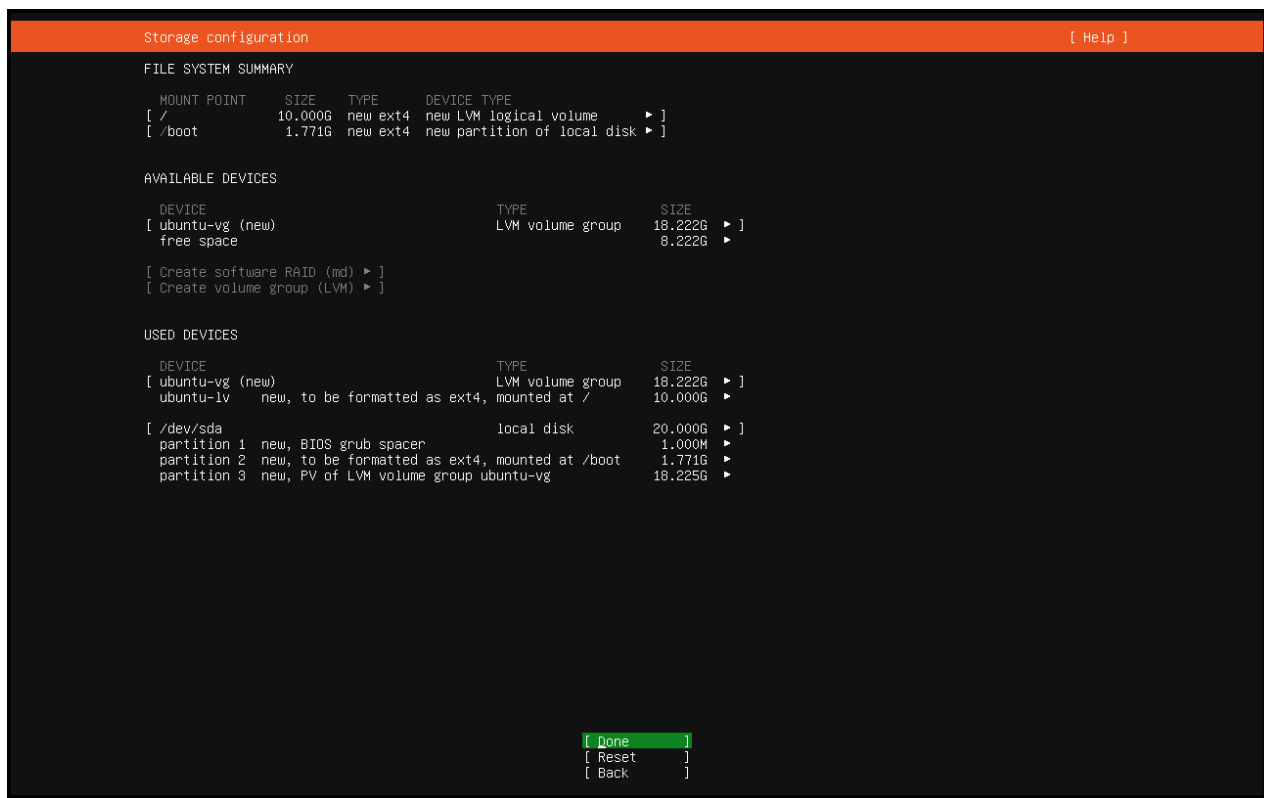
Task 10: Select the Installation Disk

22. ubuntu_selected_disk.png



Ubuntu installer showing the VMware virtual disk selected for installation.

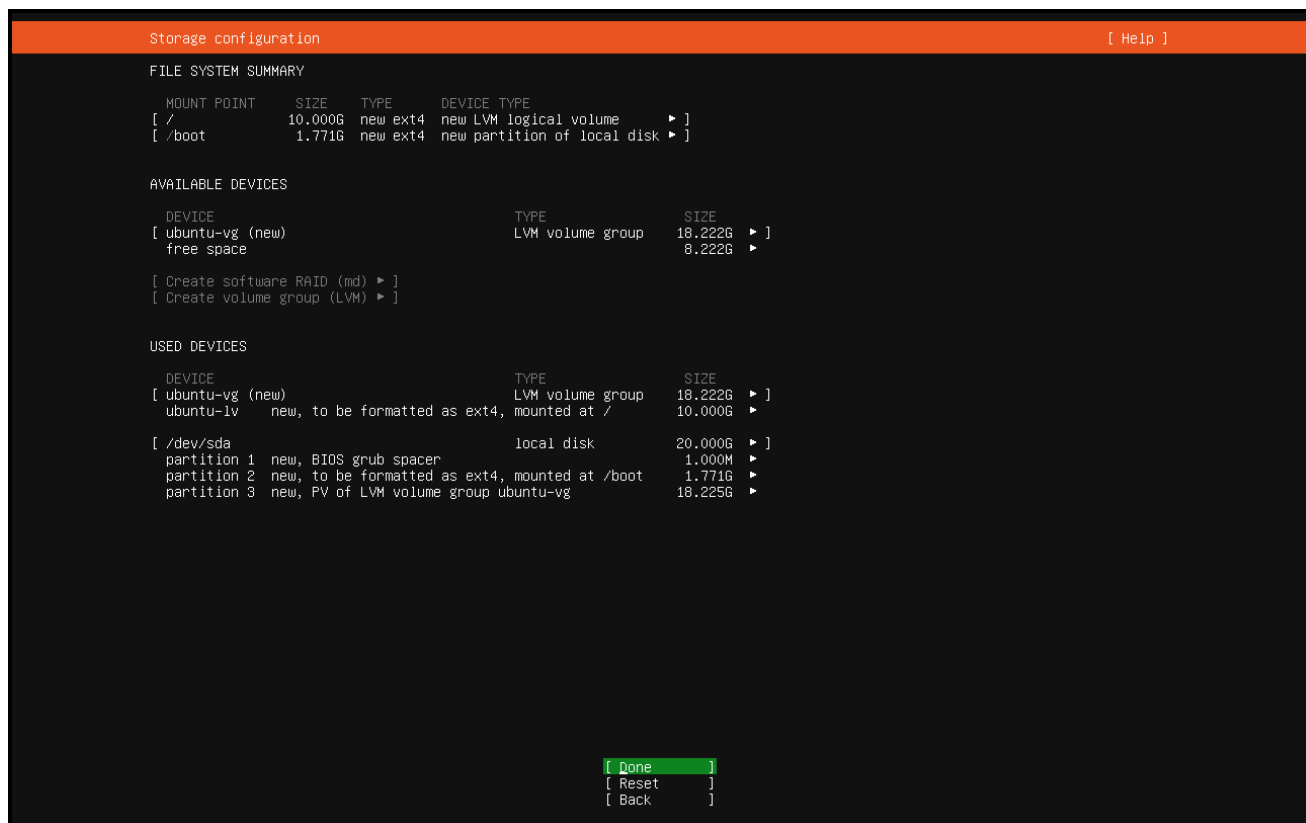
23. ubuntu_disk_details.png



Ubuntu installer showing the selected virtual disk name and size.

Task 11: Review the Partitions

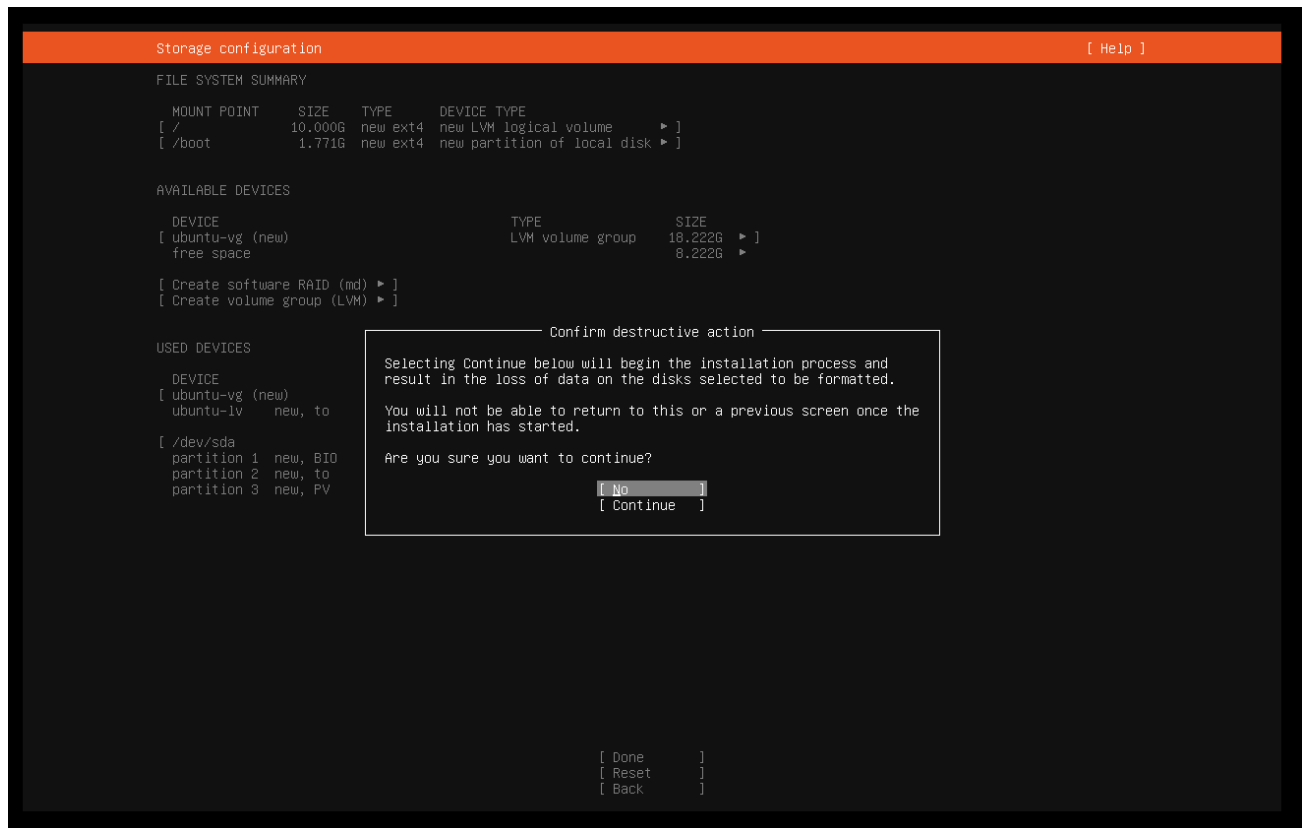
24. ubuntu_partition_layout.png



Ubuntu installer showing the proposed partition layout on the VMware virtual disk.

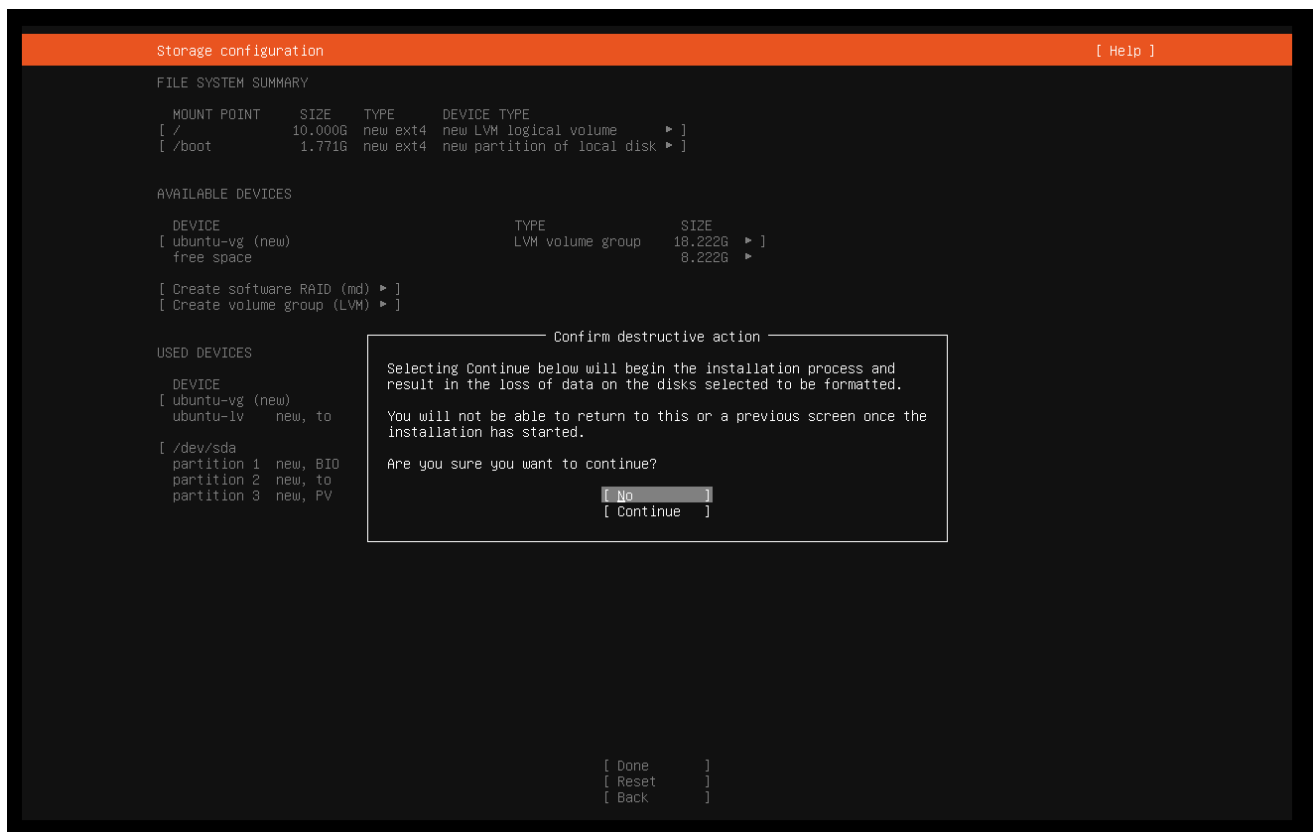
Task 12: Confirm the Storage Changes

25. ubuntu_storage_warning.png



Ubuntu installer displaying the warning about applying the storage changes.

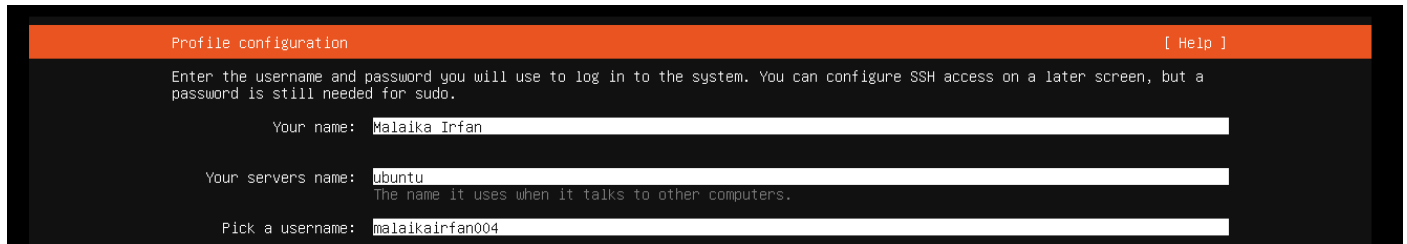
26. ubuntu_storage_confirmed.png



Ubuntu installer showing confirmation of the selected storage changes.

Task 13: Configure the Ubuntu Profile

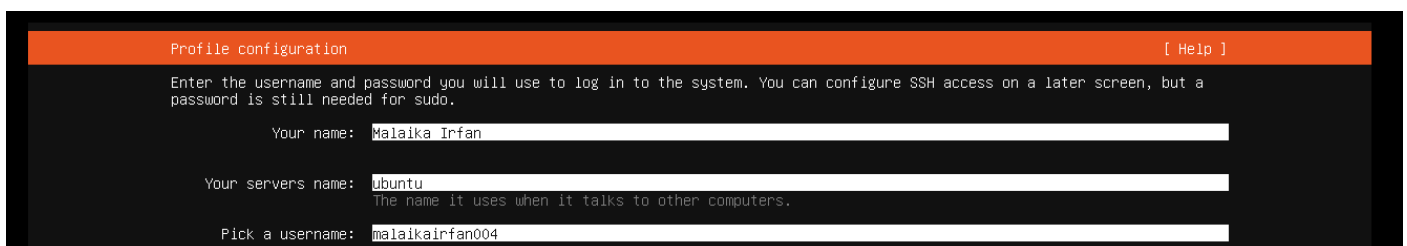
27. ubuntu_actual_name.png



The screenshot shows the 'Profile configuration' window with an orange header bar containing '[Help]'. Below the header, a message reads: 'Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.' There are three input fields: 'Your name:' with the value 'Malaika Irfan', 'Your servers name:' with the value 'ubuntu' (with a subtext 'The name it uses when it talks to other computers.'), and 'Pick a username:' with the value 'malaikairfan004'.

Ubuntu profile setup showing the user's complete actual name.

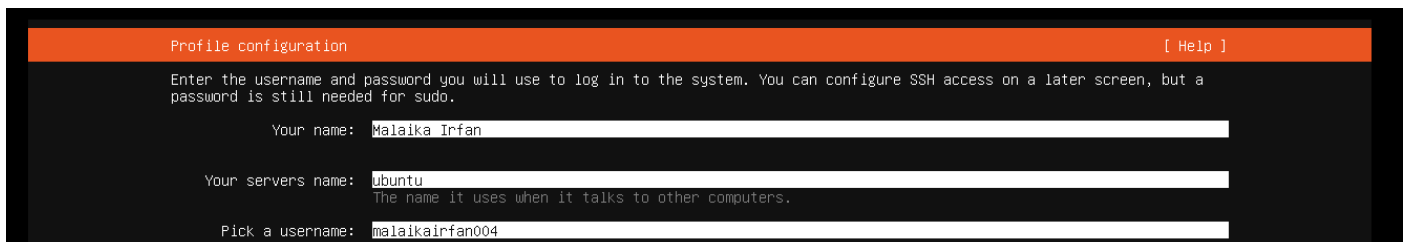
28. ubuntu_server_name.png



This screenshot is identical to the previous one, showing the 'Profile configuration' window with the same fields and values: 'Your name: Malaika Irfan', 'Your servers name: ubuntu', and 'Pick a username: malaikairfan004'.

Ubuntu profile setup showing the server name configured as ubuntu.

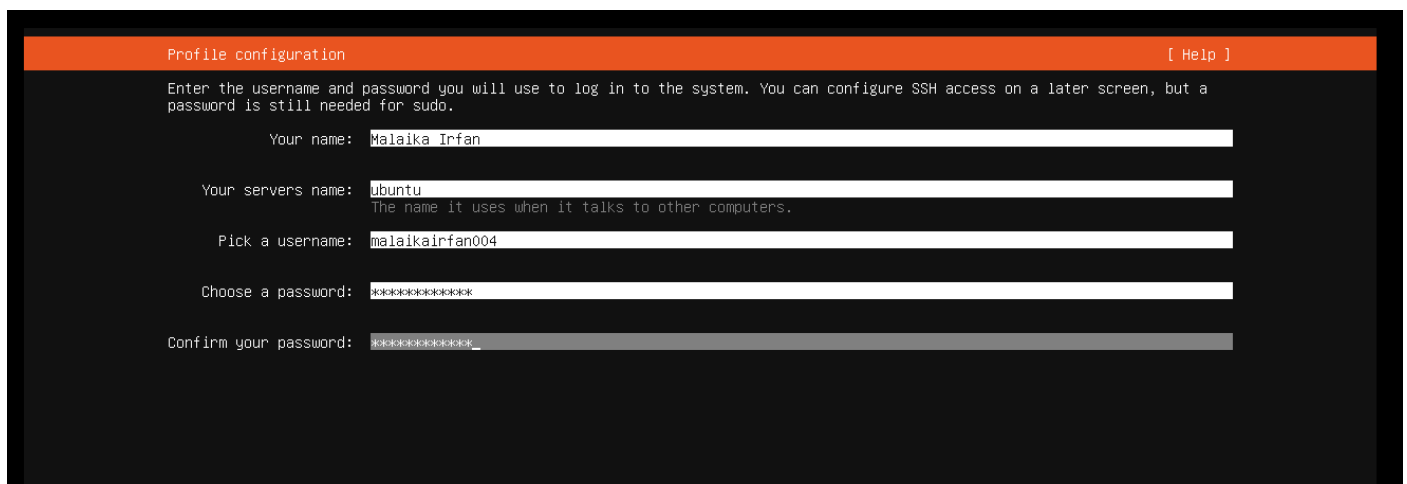
29. ubuntu_username.png



This screenshot is identical to the previous ones, showing the 'Profile configuration' window with the same fields and values: 'Your name: Malaika Irfan', 'Your servers name: ubuntu', and 'Pick a username: malaikairfan004'.

Ubuntu profile setup showing the GitHub username configured as the Ubuntu username.

30. ubuntu_profile_setup.png

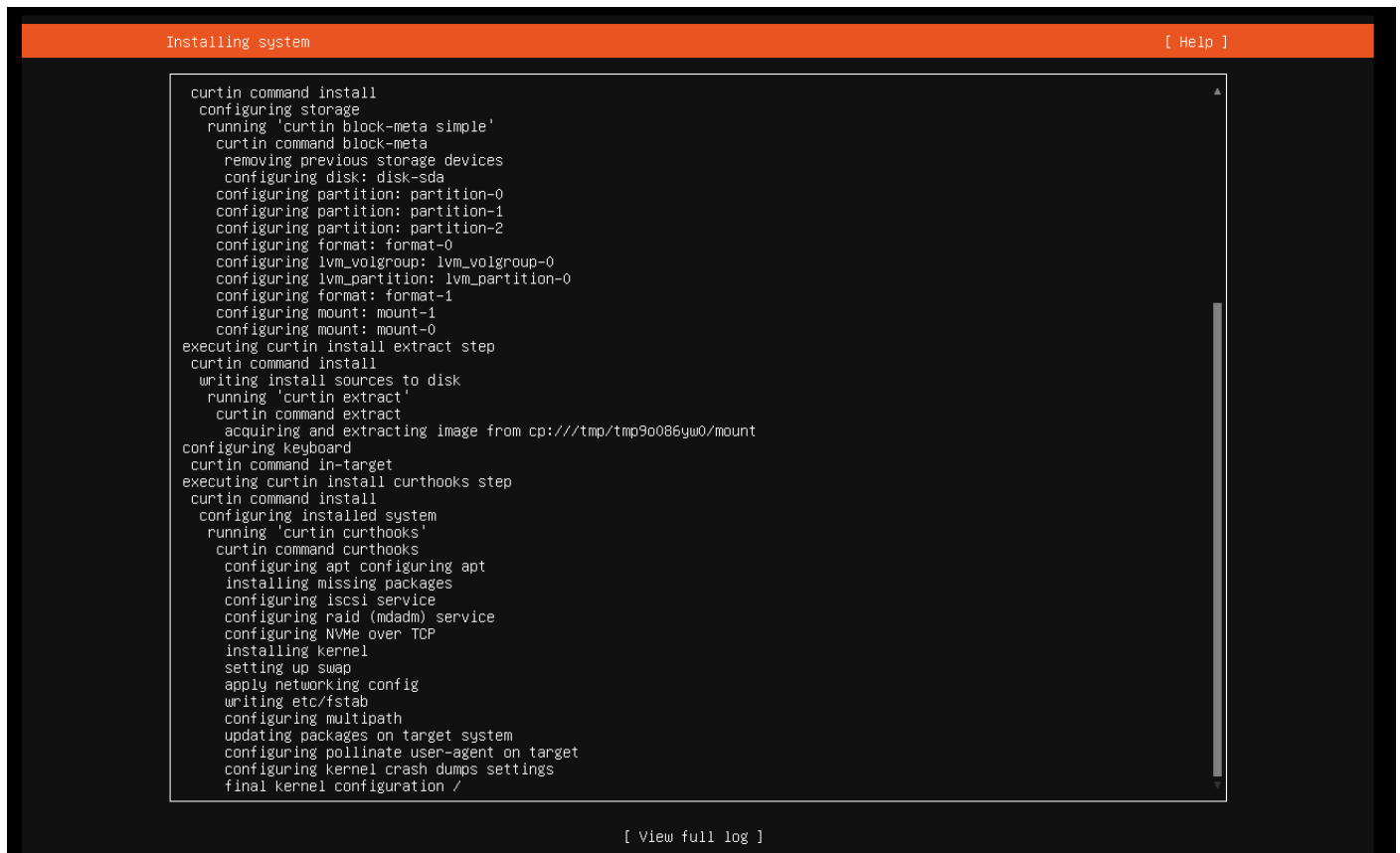


The screenshot shows the 'Profile configuration' window with the same fields as before, but with an additional field at the bottom: 'Choose a password:' with the value '*****'. The 'Confirm your password:' field is also present with the value '*****'.

Ubuntu profile setup showing the completed account information before installation.

Task 14: Complete the Software Installation

31. ubuntu_installation_progress.png



The screenshot shows the Ubuntu Server installer window titled "Installing system" with a "[Help]" button in the top right corner. The main area displays a list of installation steps in a terminal-like font. The steps include configuring storage, running 'curtin block-meta simple', configuring disk, partitions, and formats, executing the 'curtin install extract' step, configuring keyboard, running 'curtin in-target', executing the 'curtin install curthooks' step, and finally configuring the installed system, including apt, missing packages, iSCSI, RAID, NVMe, TCP, kernel, swap, networking, fstab, multipath, and updating packages. At the bottom, there is a "[View full log]" button.

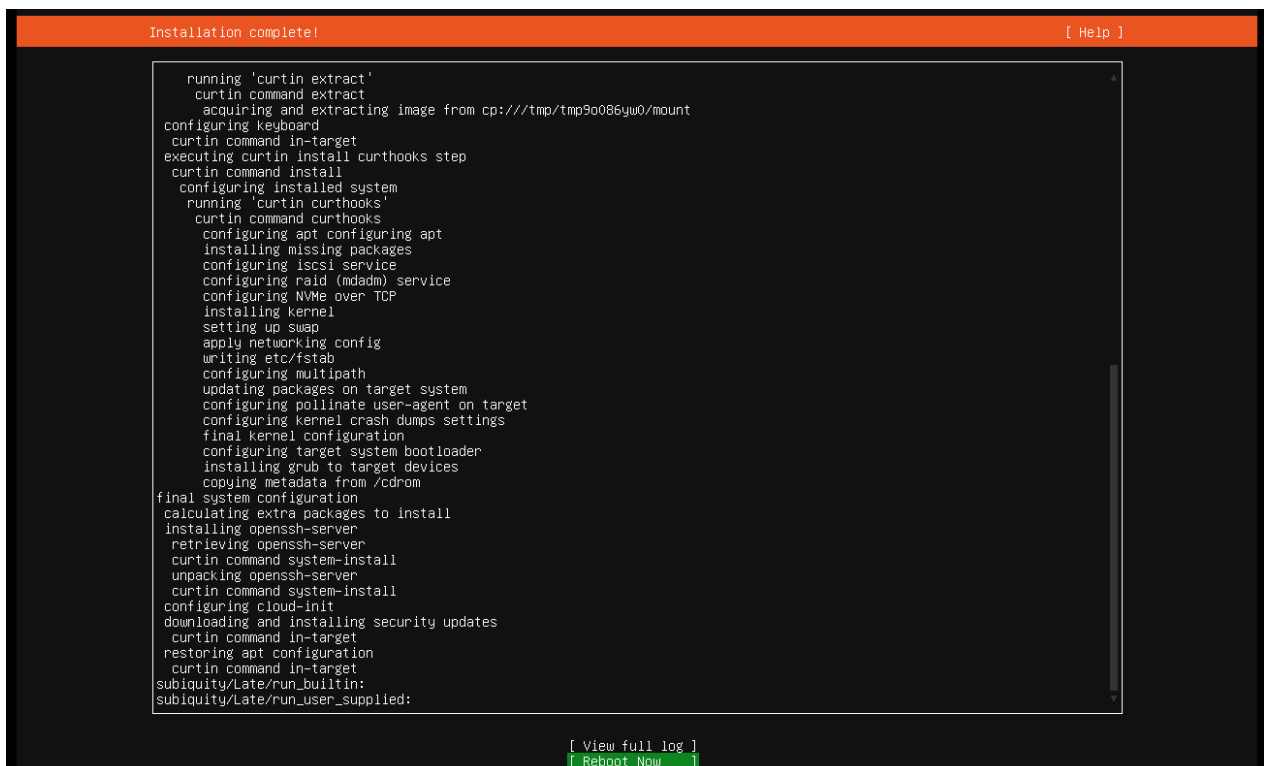
```
curtin command install
configuring storage
  running 'curtin block-meta simple'
curtin command block-meta
  removing previous storage devices
  configuring disk: disk-sda
  configuring partition: partition-0
  configuring partition: partition-1
  configuring partition: partition-2
  configuring format: format-0
  configuring lvm_volgroup: lvm_volgroup-0
  configuring lvm_partition: lvm_partition-0
  configuring format: format-1
  configuring mount: mount-1
  configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
curtin command extract
  acquiring and extracting image from cp:///tmp/tmp9o086yw0/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
curtin command curthooks
  configuring apt configuring apt
  installing missing packages
  configuring iSCSI service
  configuring raid (mdadm) service
  configuring NVMe over TCP
  installing kernel
  setting up swap
  apply networking config
  writing etc/fstab
  configuring multipath
  updating packages on target system
  configuring pollinate user-agent on target
  configuring kernel crash dumps settings
  final kernel configuration /
```

[View full log]

Ubuntu Server installer showing the software installation in progress.

Task 15: Reboot and Verify the Ubuntu Identity

32. ubuntu_installation_complete.png



The screenshot shows the Ubuntu Server installer window titled "Installation complete!" with a "[Help]" button in the top right corner. The main area displays a list of installation steps in a terminal-like font. The steps include running 'curtin extract', curtin command extract, acquiring and extracting image from cp:///tmp/tmp9o086yw0/mount, configuring keyboard, curtin command in-target, executing the 'curtin install curthooks' step, curtin command install, configuring the installed system, running 'curtin curthooks', curtin command curthooks, configuring apt, installing missing packages, configuring iSCSI, RAID, NVMe, TCP, kernel, swap, networking, fstab, multipath, updating packages, configuring pollinate user-agent, configuring kernel crash dumps, final kernel configuration, configuring target system bootloader, installing grub, copying metadata, final system configuration, calculating extra packages, installing openssh-server, retrieving openssh-server, curtin command system-install, unpacking openssh-server, curtin command system-install, configuring cloud-init, downloading and installing security updates, curtin command in-target, restoring apt configuration, curtin command in-target, subiquity/Late/run_builtin, and subiquity/Late/run_user_supplied. At the bottom, there are two buttons: "[View full log]" and "[Reboot Now]".

```
running 'curtin extract'
curtin command extract
  acquiring and extracting image from cp:///tmp/tmp9o086yw0/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
curtin command curthooks
  configuring apt configuring apt
  installing missing packages
  configuring iSCSI service
  configuring raid (mdadm) service
  configuring NVMe over TCP
  installing kernel
  setting up swap
  apply networking config
  writing etc/fstab
  configuring multipath
  updating packages on target system
  configuring pollinate user-agent on target
  configuring kernel crash dumps settings
  final kernel configuration
  configuring target system bootloader
  installing grub to target devices
  copying metadata from /cdrom
final system configuration
  calculating extra packages to install
  installing openssh-server
  retrieving openssh-server
  curtin command system-install
  unpacking openssh-server
  curtin command system-install
  configuring cloud-init
  downloading and installing security updates
  curtin command in-target
  restoring apt configuration
  curtin command in-target
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:
```

[View full log]
[Reboot Now]

Ubuntu Server installer showing that the installation has completed successfully.

33. ubuntu_login_screen.png

```
Ubuntu 26.04.1 LTS ubuntu tty1

Try contacting this VM's SSH server via 'ssh vsock%3216925328' from host.

ubuntu login:
```

Installed Ubuntu Server displaying the login screen.

34. ubuntu_terminal_login.png

```
Ubuntu 26.04.1 LTS ubuntu tty1

Try contacting this VM's SSH server via 'ssh vsock%3216925328' from host.

ubuntu login: malaikairfan004
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Mon Sep 14 02:56:35 PM UTC 2026

   System load:  0.0               Processes:             215
   Usage of /:   50.5% of 9.75GB    Users logged in:       0
   Memory usage: 21%               IPv4 address for ens33: 192.168.138.129
   Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

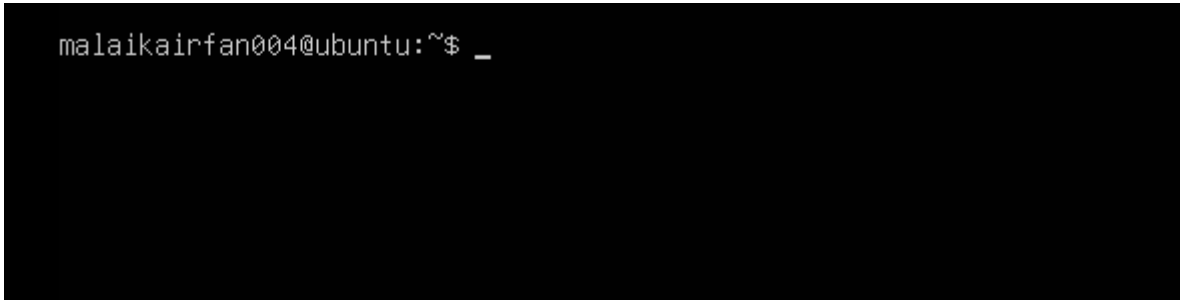
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

malaikairfan004@ubuntu:~$ _
```

Ubuntu terminal showing a successful login with the configured username.

35. ubuntu_identity_verified.png

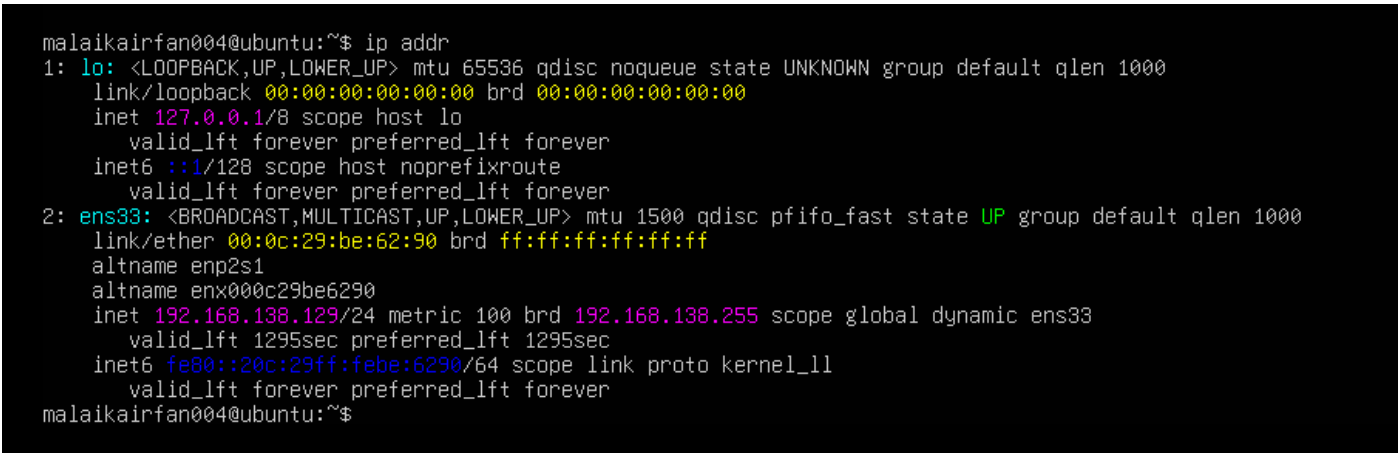


```
malaikairfan004@ubuntu:~$ _
```

Ubuntu terminal showing the verified malaikairfan004@ubuntu:~\$ prompt.

Task 16: Find the Ubuntu Server IP Address

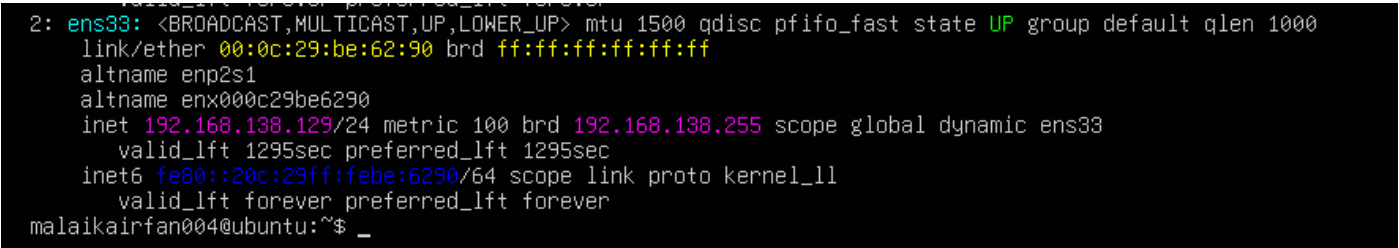
36. ubuntu_ip_addr_command.png



```
malaikairfan004@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:be:62:90 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29be6290
    inet 192.168.138.129/24 metric 100 brd 192.168.138.255 scope global dynamic ens33
        valid_lft 1295sec preferred_lft 1295sec
    inet6 fe80::20c:29ff:febe:6290/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
malaikairfan004@ubuntu:~$
```

Ubuntu terminal showing the ip addr command and its network interface output.

37. ubuntu_ip_address.png

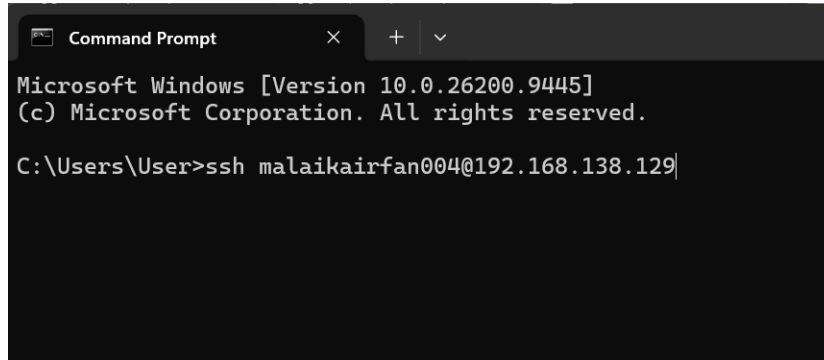


```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:be:62:90 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29be6290
    inet 192.168.138.129/24 metric 100 brd 192.168.138.255 scope global dynamic ens33
        valid_lft 1295sec preferred_lft 1295sec
    inet6 fe80::20c:29ff:febe:6290/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
malaikairfan004@ubuntu:~$ _
```

Ubuntu terminal clearly showing the active interface's IPv4 address.

Task 17: Connect to Ubuntu Server from Windows

38. windows_ssh_command.png

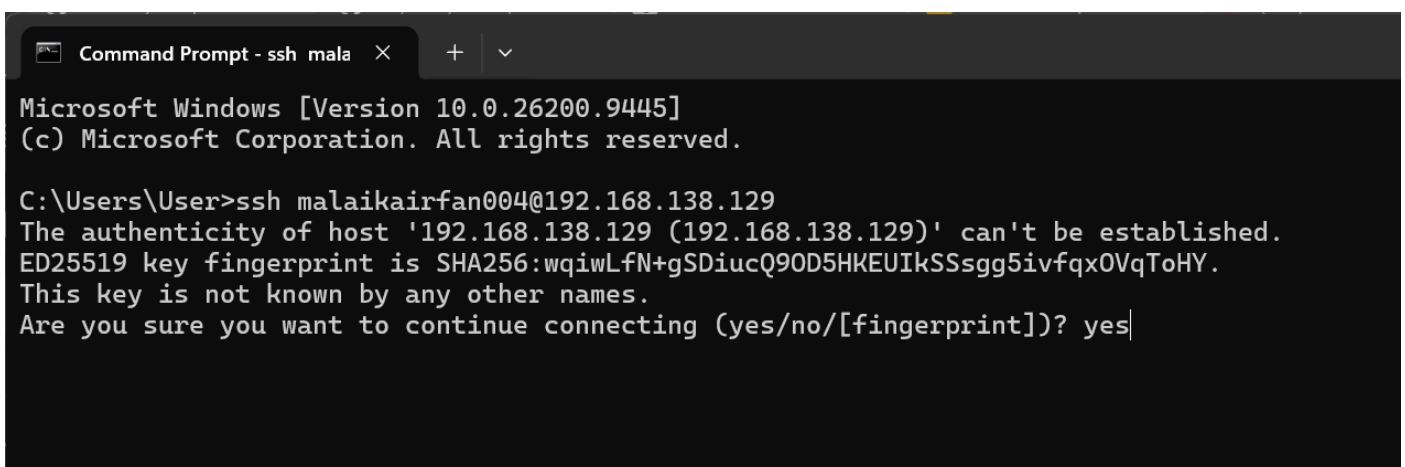


```
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh malaikairfan004@192.168.138.129
```

Windows terminal showing the SSH command used to connect to Ubuntu Server.

39. windows_ssh_fingerprint.png

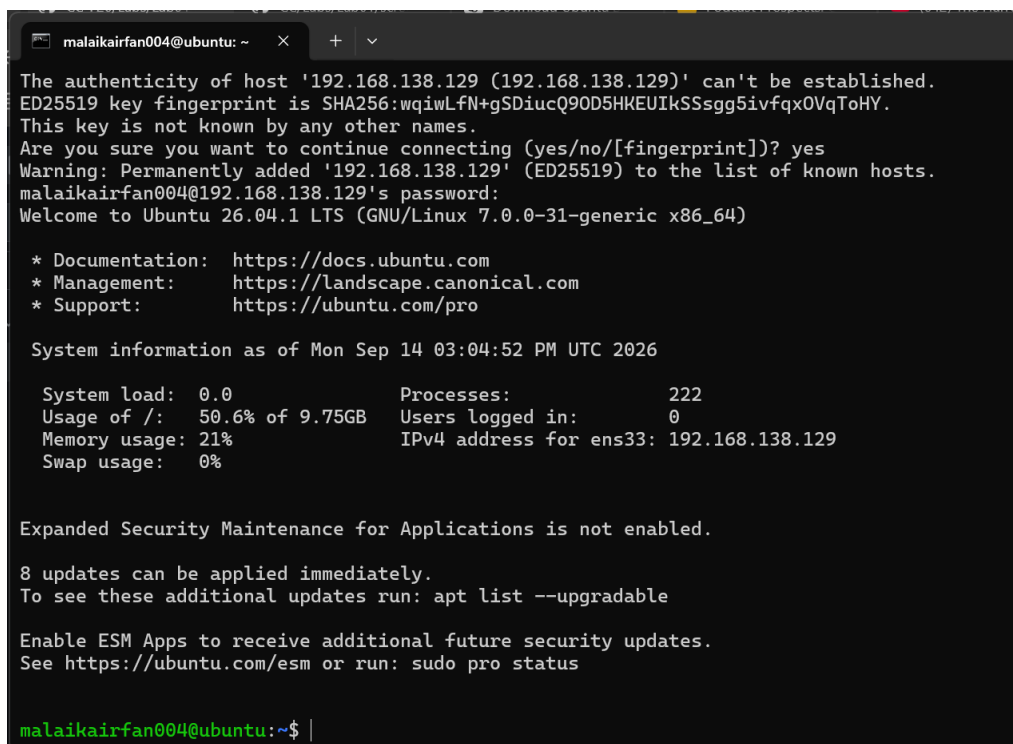


```
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh malaikairfan004@192.168.138.129
The authenticity of host '192.168.138.129 (192.168.138.129)' can't be established.
ED25519 key fingerprint is SHA256:wqiwlFn+gSDiucQ90D5HKEUIkSSsgg5ivfqx0VqToHY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

Windows terminal showing acceptance of the Ubuntu Server SSH fingerprint.

40. windows_ssh_login.png



```
malaikairfan004@ubuntu: ~
The authenticity of host '192.168.138.129 (192.168.138.129)' can't be established.
ED25519 key fingerprint is SHA256:wqiwlFn+gSDiucQ90D5HKEUIkSSsgg5ivfqx0VqToHY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.138.129' (ED25519) to the list of known hosts.
malaikairfan004@192.168.138.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Mon Sep 14 03:04:52 PM UTC 2026

System load: 0.0          Processes:                222
Usage of /:  50.6% of 9.75GB Users logged in:          0
Memory usage: 21%         IPv4 address for ens33: 192.168.138.129
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

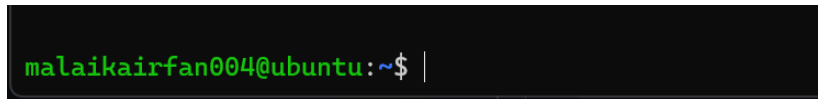
8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

malaikairfan004@ubuntu:~$
```

Windows terminal showing a successful SSH login to the Ubuntu Server.

41. windows_ssh_identity.png



Windows terminal showing the remote malaikairfan004@ubuntu identity after SSH login.